

# Predicting Sediment Particle Size Distributions Using Machine Learning

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## Objective

To predict two key sediment particle-size distribution parameters - median particle diameter ( $d_{50}$ ) and distribution variance ( $\sigma^2$ ) - from environmental, optical and physical measurements using machine-learning regression models.

## Project Overview

Accurate sediment particle-size estimates are useful for understanding sediment transport, coastal erosion, contaminant movement and related marine processes. This project implements a machine-learning workflow inspired by prior research to model the nonlinear relationships between measurable water-column conditions and suspended sediment characteristics.

## Dataset and Preprocessing

The original South San Francisco Bay field dataset was not publicly available, so the implementation uses a synthetic dataset of 1,648 hourly samples. Eight predictors are included: salinity (S), wave-orbital velocity (ub), particle refractive index (np), water temperature (T), two optical absorption ratios ( $a_{676}/a_{650}$  and  $a_{450}/a_{676}$ ), chlorophyll-a (chl\_a), and flow velocity (u). Missing values and extreme target outliers are removed, followed by a 70% training / 30% testing split.

## Methodology

Two regression approaches are compared. Random Forest Regression uses all eight input features and 400 trees, with out-of-bag evaluation during model selection. Support Vector Regression (RBF kernel) uses the four most important features - S, ub, np and T - and is tuned using 5-fold GridSearchCV over C and epsilon. Performance is evaluated using the coefficient of determination ( $R^2$ ).

## Results

| Model                | $R^2$ for $d_{50}$ | $R^2$ for $\sigma^2$ |
|----------------------|--------------------|----------------------|
| Random Forest        | approx. 0.84       | approx. 0.91         |
| SVR (top 4 features) | approx. 0.84       | approx. 0.90         |

## Key Findings and Conclusion

Both models achieve strong predictive performance, indicating that the selected environmental and optical variables contain useful information for estimating sediment particle-size distributions. Random Forest gives the strongest  $\sigma^2$  result, while SVR reaches nearly the same accuracy with only four features, making it a more compact model. The main limitation is the use of synthetic data; future work should validate the workflow on real field measurements and assess generalization across locations and seasons.

**Reference:** G. Egan, "Machine Learning for Predicting Sediment Particle Size Distributions," Stanford CS229 project.  
Project repository: [github.com/Siddhesh249/Predicting-Sediment-Particle-Size-Distributions](https://github.com/Siddhesh249/Predicting-Sediment-Particle-Size-Distributions)